物理模型推导题目

Theoretical Physics Derivation Problems

GRS

2026 年 3 月

重要的 *200* 级公式都要记住。

————Spider

Part 1 力学

Problem 1

证明抛体运动包络线方程:

$$y = \frac{v_0^2}{2g} - \frac{g}{2v_0^2}x^2.$$

Problem 2

证明曲率半径:

$$\rho = \frac{(1+y'^2)^{3/2}}{|y''|} = \frac{1}{\kappa}.$$

Problem 3

证明软绳有约束:

$$\frac{\partial v_c}{\partial s} + \frac{v_n}{\rho} = 0.$$

Problem 4

证明 $(m, M, Wall)$ 系统中 $M = km$ 物块与 m 物块在 $k \gg 1$ 时，碰撞次数:

$$n = \lfloor \sqrt{k}\pi \rfloor.$$

Problem 5

导出平方反比力轨道:

$$r = \frac{\dfrac{L^2}{GMm^2}}{1 + \sqrt{1 + \dfrac{2EL^2}{G^2M^2m^3}}\cos\theta} = \frac{p}{1 + e\cos\theta}.$$

Problem 6

导出立方反比力轨道，分 $E>0, e>1$, $E=0, e=1$, $E<0, e<1$ 的排列组合讨论.

Problem 7

求 $V=-\dfrac{GMm}{r}+\dfrac{1}{2}m\alpha r^2, \quad \alpha \ll 1$ 微扰下的进动角速度.

Problem 8

求 $V=-\dfrac{GMm}{r}+\dfrac{\beta}{r^{1+k}}$ 微扰下的进动角速度.

Problem 9

长 l 的曲杆，铰接于点 $A(x_0,0,0)$ 处，另一端在 $B(0,y_0,f(y_0))$ 处与钢丝接触。平衡的最小值为:

$$\mu_m=\frac{x_0 f'(y_0)-y_0}{x_0\sqrt{x_0^2+y_0^2+f'(y_0)^2}}.$$

Problem 10

导出悬链线方程:

$$y=\frac{T_0}{\lambda g}\left(\cosh\left(\frac{\lambda g}{T_0}x\right)-1\right).$$

Problem 11

导出引力场悬链线:

$$r = -\frac{\dfrac{1-A}{A} r_0}{1 - \dfrac{1}{A}\cos(\sqrt{1-A^2}\theta)}.$$

Problem 12

导出卢瑟福散射在质心系下的微分截面:

$$\frac{d\sigma}{d\Omega} = \left(\frac{1}{4\pi\epsilon_0}\frac{Z_1 Z_2 e^2}{4E}\right)^2 \frac{1}{\sin^4(\theta_c/2)}.$$

Problem 13

导出卢瑟福散射在实验室系下的微分截面:

$$\frac{d\sigma}{d\Omega} = \left(\frac{1}{4\pi\epsilon_0}\frac{Z_1 Z_2 e^2}{2E_1}\right)^2 \frac{\left[\cos\theta_L + \sqrt{1 - \left(\dfrac{m_1}{m_2}\sin\theta_L\right)^2}\right]^2}{\sqrt{1 - \left(\dfrac{m_1}{m_2}\sin\theta_L\right)^2}\sin^4\theta_L}.$$

Problem 14

导出欧拉运动学方程:

$$\omega_1 = \dot{\phi}\sin\theta\sin\psi + \dot{\theta}\cos\psi,$$

$$\omega_2 = \dot{\phi}\sin\theta\cos\psi - \dot{\theta}\sin\psi,$$

$$\omega_3 = \dot{\phi}\cos\theta + \dot{\psi}.$$

Problem 15

导出欧拉动力学方程:

$$I_1\dot{\omega}_i - (I_j - I_k)\omega_j\omega_k = M_i, \quad (i,j,k)\text{轮换}.$$

Problem 16

导出轴对称陀螺在 $\vec{\Pi}$ 空间（角动量空间）的轨道:

$$\Pi_1^2 + \Pi_2^2 + \Pi_3^2 = L^2,$$

$$\frac{\Pi_1^2}{2I_1T} + \frac{\Pi_2^2}{2I_2T} + \frac{\Pi_3^2}{2I_3T} = 1.$$

Problem 17

导出自由不对称陀螺的角速度:

$$\Omega_1(\tau) = \sqrt{\frac{2EI_1 - L^2}{I_1(I_3 - I_1)}}\mathrm{cn}(\tau),$$

$$\Omega_2(\tau) = \sqrt{\frac{2EI_2 - L^2}{I_2(I_3 - I_2)}}\mathrm{sn}(\tau),$$

$$\Omega_3(\tau) = \sqrt{\frac{L^2 - 2EI_3}{I_3(I_3 - I_1)}}\mathrm{dn}(\tau).$$

Problem 18

证明对称陀螺定点转动:

$$\dot{\phi} = \frac{L_3 - L_2\cos\theta}{I_1'\sin^2\theta},$$

$$\dot{\psi} = \frac{L_2}{I_3} - \frac{L_3 - L_2\cos\theta}{I_1'\sin^2\theta}\cos\theta,$$

$$\text{s.t. } I_1' = I_1 + ml^2.$$

Problem 19

证明刚体所有运动构成的流形 Q 有:

$$\mathbb{Q} = \mathbb{R}^3 \rtimes SO(3).$$

Problem 20

找到天体运动拉格朗日点 $L_i, i = 1, 2, 3, 4, 5$，并讨论运动模式.

Problem 21

导出平面双振子在 $\phi_1(0) = -\phi_2(0) = \phi_0 \ll 1$, $\dot{\phi}_1(0) = \dot{\phi}_2(0) = 0$ 初态下系统的解

$$\phi_1(t) = \frac{\phi_0}{4} \begin{pmatrix} (2+\sqrt{2})\cos(\omega_1 t) + (2-\sqrt{2})\cos(\omega_2 t) \\ -(2+2\sqrt{2})\cos(\omega_1 t) + (2\sqrt{2}-2)\cos(\omega_2 t) \end{pmatrix},$$

$$\text{s.t. } \omega_{1,2} = \sqrt{\frac{k}{m}(2 \pm \sqrt{2})}.$$

Problem 22

导出成环原子链的本征频率:

$$\omega^2(p) = 2\omega_0^2(1 - \cos(p)),$$

$$p = \frac{2\pi l}{N}, \quad l \in \left[-\frac{N}{2}, \frac{N}{2}\right].$$

Problem 23

导出单摆周期的修正:

$$T = 2\pi\sqrt{\frac{l}{g}}\left(1 + \frac{1}{16}\theta_0^2 + \cdots\right).$$

Problem 24

对参数振子: $\omega^2(t) = \omega_0^2\big(1 + h\cos(\gamma t)\big)$, $h \ll 1$, $\gamma = 2\omega_0 + 2\varepsilon$, $\varepsilon \ll \omega_0$，存在指数增长解的条件:

$$|\varepsilon| < \frac{1}{4}h\omega_0.$$

Problem 25

导出二维欧氏平面上在 $V = \frac{1}{2}k(x^2 + y^2)$ 与 $\vec{F} = -m\vec{\Omega} \times \vec{v}$ 下质点 m 的运动模式.

Problem 26

导出一维双原子链 m, M 的色散关系:

$$\omega_{\pm}^2 = \frac{k(M+m)}{Mm}\left\{1 \pm \sqrt{1 - \frac{4Mm\sin^2(\frac{1}{2}ka)}{(M+m)^2}}\right\}.$$

Problem 27

导出 N-C 方程:

$$\rho\frac{d^2\vec{\xi}}{dt^2} = (\lambda + \mu)\nabla(\nabla \cdot \vec{\xi}) + \mu\nabla^2\vec{\xi}.$$

Problem 28

导出弹性波波速:

$$c_{//} = \sqrt{\frac{\lambda + 2\mu}{\rho}}, \quad c_{\perp} = \sqrt{\frac{\mu}{\rho}}.$$

Problem 29

证明:

$$\nu = \frac{\lambda}{2(\lambda + \mu)},$$
$$E = \frac{\mu(3\lambda + 2\mu)}{\lambda + \mu},$$
$$\lambda = \frac{E\nu}{(1+\nu)(1-2\nu)},$$
$$\mu = \frac{E}{2(1+\nu)}.$$

Problem 30

导出两边固定、中间拱起的轻绳形状近似解:

$$y = \frac{L\Delta}{\pi}\left(1 - \cos\left(\frac{2\pi x}{L}\right)\right).$$

Problem 31

质量 m 的弹簧振子以 ω 频率振动时的机械能:

$$x = \frac{\sin\left(\sqrt{\frac{m}{k}}\frac{\omega}{l}\xi\right)}{\sqrt{\frac{m}{k}}\frac{\omega}{l}\cos\left(\sqrt{\frac{m}{k}}\frac{\omega}{l}\right)}.$$

Problem 32

无原长 $l \ll \dfrac{mg}{k}$ 质量 m 弹簧悬链线:

$$y = -\frac{mg}{k}\left(\frac{x}{d}\right)^2 + \frac{mg}{k}\left(\frac{x}{d}\right).$$

Problem 33

导出泊肃叶公式:

$$Q = \frac{\pi R^4 \Delta p}{8\eta L}.$$

Problem 34

证明欧拉动力学方程（流体）:

$$\frac{\partial}{\partial t}(\rho\vec{v}) + \nabla\cdot(\rho\vec{v}\otimes\vec{v}) + \nabla p = 0.$$

Problem 35

证明理想流体，准静态绝热下能量守恒方程:

$$\frac{\partial}{\partial t}\left(u_s + \frac{v^2}{2}\right) + \nabla\cdot\left[\left(u_s + \frac{v^2}{2} + \frac{p}{\rho}\right)\vec{v}\right] = 0.$$

Problem 36

证明 N-S 方程:

$$\rho\left(\frac{\partial\vec{v}}{\partial t} + (\vec{v}\cdot\nabla)\vec{v}\right) = -\nabla p + \eta\nabla^2\vec{v} + \vec{f}.$$

Problem 37

讨论阻尼受迫振动的运动模式.

Problem 38

证明经典多普勒公式:

$$\nu = \nu_0 \frac{u - v_B}{u - v_S \cos\varphi_S}.$$

Problem 39

单摆，悬挂点受 $\vec{s} = \delta_0(\cos\alpha\,\vec{\imath} + \sin\alpha\,\vec{\jmath})\cos(\omega t)$ 驱动，证明微扰下低频小振动频率:

$$\Omega \approx \sqrt{\frac{g}{l}\cos\theta - \frac{\delta_0^2\cos(2\theta - 2\alpha)\omega^2}{2l^2}}.$$

Problem 40

m_1, m_2, l_1, l_2 双摆，驱动 $\delta = \delta_0\cos(\omega t)$ 求简正振动频率满足的方程: $A\omega^4 + B\omega^2 + C = 0$. 其中:

$$A = 1,$$

$$B = -\frac{m_1 + m_2}{m_1}\left[g\left(\frac{1}{l_1} + \frac{1}{l_2}\right) + \frac{(m_1 + m_2)^2(l_1 + l_2)}{2m_1 l_1^2 l_2^2}\delta_0^2\omega^2 + \frac{m_2\delta_0^2\omega^2}{m_1 l_1 l_2}\right],$$

$$C = \frac{m_1 + m_2}{m_1 l_1 l_2}g^2 + \frac{(m_1 + m_2)^2}{2m_1^2 l_1^2 l_2^2}(l_1 + l_2)\delta_0^2\omega^2 g + \frac{(m_1 + m_2)^2}{4m_1^2 l_1^2 l_2^2}\delta_0^4\omega^4.$$

Part 2 热学

Problem 41

证明 Maxwell 关系:

$$\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial p}{\partial S}\right)_V,$$
$$\left(\frac{\partial T}{\partial p}\right)_S = \left(\frac{\partial V}{\partial S}\right)_p,$$
$$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial p}{\partial T}\right)_V,$$
$$\left(\frac{\partial S}{\partial p}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_p.$$

Problem 42

证明能态、焓态定理:

$$\left(\frac{\partial U}{\partial V}\right)_T = T\left(\frac{\partial p}{\partial T}\right)_V - p,$$
$$\left(\frac{\partial H}{\partial p}\right)_T = T\left(\frac{\partial V}{\partial T}\right)_p + V.$$

Problem 43

证明响应函数间有:

$$\alpha = \beta\kappa_T p,$$
$$\frac{C_p}{C_V} = \frac{\kappa_T}{\kappa_S},$$
$$C_p - C_V = TV\alpha\beta.$$

Problem 44

证明 Gibbs-Helmholtz 方程:

$$\left(\frac{\partial(G/T)}{\partial T}\right)_p = -\frac{H}{T^2}.$$

Problem 45

证明绝热大气满足:

$$n = n_0\left(1-(1-\frac{1}{\gamma})\frac{mgz}{kT_0}\right)^{\frac{1}{\gamma-1}}.$$

Problem 46

证明等温大气满足:

$$n = n_0 \exp\left(-\frac{mgh}{kT}\right).$$

Problem 47

证明范德瓦尔斯气体节流的 J-T 系数:

$$\mu = \left(\frac{\partial T}{\partial p}\right)_H = \frac{2aV_m(V_m-b)^2 - bRTV_m^3}{R^2TV_m^3 + C_{V,m}[RTV_m^3 - 2a(V_m-b)^2]}.$$

Problem 48

证明声速公式:

$$c = \sqrt{\gamma\frac{RT}{M}}.$$

Problem 49

证明奥托循环效率:

$$\eta_{\text{Otto}} = 1 - \frac{1}{r^{\gamma-1}}, \quad r = \frac{V_1}{V_2}.$$

Problem 50

证明狄塞尔循环效率:

$$\eta_{\text{Diesel}} = 1 - \frac{\rho^{\gamma} - 1}{\gamma(\rho - 1) r^{\gamma-1}}, \quad r = \frac{V_1}{V_2}, \quad \rho = \frac{V_3}{V_2}.$$

Problem 51

证明范德瓦尔斯气体的内能和焓:

$$U = \int_{T_0}^{T} C_V dT - \frac{\nu^2 a}{V} + U_0,$$

$$H = \int_{T_0}^{T} C_V dT + \left(\frac{\nu RTV}{V - \nu b} - \frac{2\nu^2 a}{V} \right) + H_0.$$

Problem 52

证明张力系统中:

$$S_S = -A_S \left(\frac{\partial \sigma}{\partial T} \right)_{A_S}, \quad U_S = \sigma A_S - T A_S \left(\frac{\partial \sigma}{\partial T} \right)_{A_S}.$$

Problem 53

证明附加压强的 Laplace 公式:

$$\Delta p = \sigma \left(\frac{1}{R_1} + \frac{1}{R_2} \right).$$

Problem 54

证明方形容器附加高度:

$$h = \sqrt{\frac{2\sigma}{\rho g}(1 - \sin\theta)}.$$

Problem 55

证明范德瓦尔斯气体临界参数:

$$T_c = \frac{8a}{27Rb},$$

$$V_c = 3b,$$

$$p_c = \frac{a}{27b^2}.$$

Problem 56

导出饱和蒸气压方程:

$$\ln\left(\frac{P_S}{P_{S,0}}\right) = B\left(1 - \frac{T_0}{T}\right) + C\ln\left(\frac{T}{T_0}\right).$$

Problem 57

导出诱导核的中肯半径:

$$r_C = \frac{2\sigma\rho_G}{(\rho_L - \rho_G)(P' - P)}.$$

Problem 58

导出 Van't Hoff 方程:

$$\frac{d\ln K_P(T)}{dT} = \frac{\Delta H}{RT^2}.$$

Problem 59

导出正则系综的能量涨落:

$$\langle(\Delta E)^2\rangle = kT^2C_V.$$

Problem 60

导出巨正则系综的粒子数涨落和能量涨落:

$$\langle(\Delta N)^2\rangle = \frac{kT\bar{N}^2}{V}\kappa_T,$$

$$\langle(\Delta E)^2\rangle = kT^2C_V + \left(\frac{\partial\bar{E}}{\partial\bar{N}}\right)^2_{T,V}\frac{kT\bar{N}^2}{V}\kappa_T.$$

Problem 61

导出单原子分子理想气体的熵:

$$S = Nk\ln\left[\frac{V}{N}\left(\frac{2\pi mkT}{h^2}\right)^{3/2}\right] + \frac{5}{2}Nk.$$

Problem 62

导出双原子分子气体 H_2 的定容热容 C_V.

Problem 63

导出电场中双原子分子气体的电极化率:

$$P_0 = \frac{Nd_0}{V}\left(\coth(\beta d_0 E_0) - \frac{1}{\beta d_0 E_0}\right)E.$$

Problem 64

导出混合理想气体的熵和化学势:

$$S = \sum_i N_i k \ln\left[\frac{V}{N_i \lambda_i^3} I_i(T)\right] + N_i kT \frac{d\ln I_i(T)}{dT} + \frac{5}{2}N_i k,$$

$$\mu_i = \mu_i^\Theta(T) + kT\ln\frac{p_i}{p^\Theta}.$$

Problem 65

导出光子气:

$$u(\omega, T) = \frac{\hbar}{\pi^2 c^3}\frac{\omega^3}{e^{\hbar\omega/kT} - 1},$$

$$E = \sigma V T^4,$$

$$P = \frac{\sigma}{3}T^4,$$

$$G = 0,$$

$$S = \frac{4}{3}\sigma V T^3.$$

Problem 66

证明爱因斯坦热容:

$$C_V = 3Nk\frac{(\beta\hbar\omega)^2 e^{\beta\hbar\omega}}{(e^{\beta\hbar\omega} - 1)^2}.$$

Problem 67

证明德拜低温热容:

$$C_V = \frac{12\pi^4}{5} Nk \left(\frac{T}{\theta_D}\right)^3, \quad \text{s.t. } \theta_D = \frac{\hbar\omega_D}{k}.$$

Problem 68

导出居里定律:

$$\chi = \frac{C}{T}.$$

Problem 69

导出二能级系统热容:

$$C_V = Nk \left(\frac{\Delta}{kT}\right)^2 \frac{e^{\Delta/kT}}{(1+e^{\Delta/kT})^2}, \quad \Delta = 2\epsilon.$$

Problem 70

证明弱简并理想玻色气体状态方程:

$$\frac{PV}{NkT} = 1 - \frac{\zeta(5/2)}{2^{5/2}} \frac{\lambda^3}{v} - \cdots, \quad \text{s.t. } \lambda = \sqrt{\frac{h^2}{2\pi mkT}}.$$

Problem 71

证明弱简并理想玻色气体的熵:

$$S = \frac{5}{2} Nk \frac{g_{5/2}(z)}{g_{3/2}(z)} - Nk \ln z.$$

Problem 72

验证 BEC 热容偏导在 T_c 处不连续.

Problem 73

证明 $V = \frac{1}{2}m(\omega_1^2 x^2 + \omega_2^2 y^2 + \omega_3^2 z^2)$ 中的 BEC 凝聚温度:

$$T_c = \frac{\hbar\bar{\omega}N^{1/3}}{k[\zeta(3)]^{1/3}}.$$

Problem 74

证明弱简并理想费米气体状态方程:

$$\frac{PV}{NkT} = 1 + \frac{\zeta(5/2)}{2^{5/2}}\frac{\lambda^3}{v} + \cdots .$$

Problem 75

证明零温理想费米气体:

$$\epsilon_F = \frac{\hbar^2}{2m}(3\pi^2 n)^{2/3},$$

$$p_F = \hbar(3\pi^2 n)^{1/3},$$

$$E_0 = \frac{3}{5}N\epsilon_F,$$

$$P_0 = \frac{2}{5}n\epsilon_F.$$

Problem 76

证明低温理想费米气体热容:

$$C_V = \frac{\pi^2}{2} Nk \frac{T}{T_F}, \quad \text{s.t. } T_F = \frac{\epsilon_F}{k}.$$

Problem 77

证明磁场中强简并费米气体的顺磁化率:

$$\chi_P = \frac{3}{2} \frac{n\mu_B^2}{\epsilon_F}.$$

Problem 78

证明 $V = \dfrac{1}{2} m(\omega_1^2 x^2 + \omega_2^2 y^2 + \omega_3^2 z^2)$ 中的强简并费米气体热容:

$$C_V = \pi^2 Nk \left(\frac{kT}{\mu_0} \right) + \cdots.$$

Problem 79

导出平均泄流数率:

$$\Gamma = \frac{1}{4} n\bar{v}.$$

Problem 80

导出 t 维 MB 分布:

$$f(\vec{v}) = \left(\frac{m}{2\pi kT} \right)^{t/2} e^{-\frac{mv^2}{2kT}},$$

$$f(v,t) = \frac{t\pi^{t/2}}{\Gamma(\frac{t}{2}+1)} v^{t-1} \left(\frac{m}{2\pi kT} \right)^{t/2} e^{-\frac{mv^2}{2kT}}.$$

Problem 81

证明泄流升维:

$$f(v_e, t) = f(v, t+1).$$

Problem 82

导出自由程分布:

$$P(\lambda) = e^{-\lambda/\bar{\lambda}}.$$

Problem 83

导出稀薄混合气体平均自由程:

$$\lambda_1 = \frac{1}{\sqrt{2}\pi d_1^2 n_1 + \pi(d_1+d_2)^2 n_2 \sqrt{\dfrac{m_1}{16\mu}}}, \quad \lambda_2 = \frac{1}{\sqrt{2}\pi d_2^2 n_2 + \pi(d_1+d_2)^2 n_1 \sqrt{\dfrac{m_2}{16\mu}}}.$$

Problem 84

维里展开得到实际气体状态方程:

$$\frac{p}{kT} = n + B_2(T)n^2 + B_3(T)n^3 + \cdots.$$

Problem 85

在 L-J 势下得到范德瓦尔斯方程:

$$p = \frac{NkT}{V-b} - \frac{a}{V^2}.$$

Problem 86

证明物态方程:

$$pV = NkT\left[1 - \frac{n}{6kT}\int d^3\vec{r}\,(\vec{r}\cdot\nabla U(\vec{r}))g(\vec{r})\right].$$

Problem 87

导出稀薄等离子体的热态方程:

$$p = \frac{NkT}{V} - \frac{e^2}{3V^{3/2}}\left(\frac{\pi}{kT}\right)^{1/2}\left(\sum_i N_i z_i^2\right)^{3/2}.$$

Problem 88

导出平均场近似 Ising 模型临界点热容:

$$C_H = \begin{cases} 3Nk/2, & T \to T_c^- \\ 0, & T \to T_c^+ \end{cases}.$$

Problem 89

导出一维 Ising 模型配分函数:

$$Z = \left[e^{\beta J}\cosh(\beta H) + \sqrt{e^{2\beta J}\sinh^2(\beta H) + e^{-2\beta J}}\right]^N, \quad m = \frac{\sinh(\beta H)}{\sqrt{\sinh^2(\beta H) + e^{-4\beta J}}}.$$

Problem 90

导出海森堡模型临界温度:

$$T_c = \frac{qJ}{3k}.$$

Problem 91

导出玻尔兹曼积分-微分方程:

$$\frac{\partial f}{\partial t} + \nabla_r \cdot (f\dot{\vec{r}}) + \nabla_v \cdot (f\dot{\vec{v}}) = \int (f_1 f' - f_1 f) d\vec{v}_1 \Lambda d\Omega.$$

Problem 92

导出输运系数:

$$\eta = nkT\bar{\tau}, \quad \sigma = \frac{ne^2\tau_F}{m}, \quad \kappa = \frac{\pi}{3}\frac{n\tau_F}{m}kT^2.$$

Problem 93

导出非平衡热机最优化功率时的效率:

$$\eta = 1 - \sqrt{\frac{T_L}{T_h}}.$$

Problem 94

证明爱因斯坦扩散系数:

$$D = \frac{kT}{6\pi a\eta}.$$

Part 3 电磁学

Problem 95

导出场点处的电偶极矩电场:

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{3(\hat{R} \cdot \vec{p})\hat{R} - \vec{p}}{R^3}.$$

Problem 96

导出电偶极层的电场:

$$\vec{E} = \frac{\tau \nabla \Omega}{4\pi\varepsilon_0}.$$

Problem 97

导出二维电偶极矩的电场:

$$\vec{E} = \frac{1}{2\pi\epsilon_0} \frac{2(\hat{R} \cdot \vec{p})\hat{R} - \vec{p}}{R^3}.$$

Problem 98

二维电偶极层的电场:

$$\vec{E} = \frac{\tau \nabla \theta}{2\pi\varepsilon_0}.$$

Problem 99

导出有限长带电细棒的电场:

$$\vec{E} = \frac{\lambda}{2\pi\epsilon_0 \sinh\mu} \frac{\hat{\mu}}{c\sqrt{\sinh^2\mu + \sin^2\nu}},$$

s.t. $x = c\cosh\mu\cos\nu, \quad y = c\sinh\mu\sin\nu.$

Problem 100

证明点电荷对导体球的电像:

$$q' = -\frac{R}{r}q, \quad r' = \frac{R^2}{r}.$$

Problem 101

证明线电荷对导体圆柱的电像:

$$\lambda' = -\lambda, \quad r' = \frac{R^2}{d}.$$

Problem 102

证明不带电导体球在均匀外场中的偶极矩:

$$\vec{p} = 4\pi\epsilon_0 R^3 \vec{E}_0.$$

Problem 103

导出旋转椭球体电容:

$$C = \begin{cases} \dfrac{4\pi\epsilon_0\sqrt{a^2 - b^2}}{\operatorname{arccosh}(a/b)}, & a > b \\ \dfrac{4\pi\epsilon_0\sqrt{b^2 - a^2}}{\arccos(a/b)}, & a < b \end{cases}.$$

Problem 104

导出两相同导体球电容:

$$C = 8\pi\epsilon_0 R \ln 2.$$

Problem 105

导出圆柱与无限大平板电容:

$$C = \frac{2\pi\epsilon_0 L}{\operatorname{arccosh}(d/r)}.$$

Problem 106

导出球与无限大平板电容:

$$C = 8\pi\epsilon_0\sqrt{d^2 - R^2}\sum_n \frac{R^n}{\left(d + \sqrt{d^2 - R^2}\right)^n - \left(d - \sqrt{d^2 - R^2}\right)^n}.$$

Problem 107

导出两无限长平行圆柱电容:

$$C = \frac{2\pi\epsilon_0 L}{\operatorname{arccosh}\left(\dfrac{d^2 - R_1^2 - R_2^2}{2R_1R_2}\right)}, \quad (d > |R_1 - R_2|).$$

Problem 108

导出垂直电场中的无限长介质柱极化:

$$\vec{P} = \frac{\epsilon_r - 1}{\epsilon_r + 1} 2\epsilon_0 \vec{E}_0.$$

Problem 109

导出均匀外场中的介质球极化:

$$\vec{P} = \frac{\epsilon_r - 1}{\epsilon_r + 2} 3\epsilon_0 \vec{E}_0.$$

Problem 110

导出大量均匀介质球的等效极化率:

$$\epsilon_{\text{eff}} = 1 + n\frac{\epsilon_r - 1}{2\epsilon_r + 1} 4\pi R^3.$$

Problem 111

导出磁偶极层的磁场:

$$\vec{B} = \frac{\mu_0 I}{4\pi} \nabla \Omega.$$

Problem 112

证明磁矩在时空缓变电磁场下守恒:

$$\mu = \text{const}.$$

Problem 113

证明在 $\vec{B} = (0, 0, B)$, $\vec{E} = (0, E, 0)$, $\vec{f} = -\eta\vec{v}$, $\vec{F} = -k\vec{r}$, $\vec{r}_0 = (0, 0, 0)$, $\vec{v}_0 = (v_{x0}, v_{y0}, 0)$ 条件下有:

$$\begin{aligned}\bar{r}(t) = &-\frac{iEq}{k} e^{\left(\frac{iBq+\eta}{2m}\right)t} \frac{e^{\sqrt{\left(\frac{iBq+\eta}{2m}\right)^2 - \frac{k}{m}}t} + e^{-\sqrt{\left(\frac{iBq+\eta}{2m}\right)^2 - \frac{k}{m}}t}}{2} \\ &+ \frac{v_{x0} + iv_{y0} - \frac{iBq+\eta}{2m} \cdot \frac{iEq}{k}}{\sqrt{\left(\frac{iBq+\eta}{2m}\right)^2 - \frac{k}{m}}} e^{\left(\frac{iBq+\eta}{2m}\right)t} \frac{e^{\sqrt{\left(\frac{iBq+\eta}{2m}\right)^2 - \frac{k}{m}}t} - e^{-\sqrt{\left(\frac{iBq+\eta}{2m}\right)^2 - \frac{k}{m}}t}}{2} + \frac{iEq}{k}.\end{aligned}$$

Problem 114

证明 Fresnel 公式:

$$\left(\frac{E_{0r}}{E_{0i}}\right)_s = \frac{n_1\cos\theta_i - n_2\cos\theta_t}{n_1\cos\theta_i + n_2\cos\theta_t}, \quad \left(\frac{E_{0t}}{E_{0i}}\right)_s = \frac{2n_1\cos\theta_i}{n_1\cos\theta_i + n_2\cos\theta_t},$$

$$\left(\frac{E_{0r}}{E_{0i}}\right)_p = \frac{n_2\cos\theta_i - n_1\cos\theta_t}{n_2\cos\theta_i + n_1\cos\theta_t}, \quad \left(\frac{E_{0t}}{E_{0i}}\right)_p = \frac{2n_1\cos\theta_i}{n_2\cos\theta_i + n_1\cos\theta_t}.$$

Problem 115

导出隐失波沿界面的能流密度:

$$S_x = \frac{1}{2}\sqrt{\frac{\epsilon_0}{\mu_0}}\frac{\sin\theta}{\sin\theta_c}|E_0''|^2 e^{-2\kappa z}.$$

Problem 116

导出一维光子晶体 s, p 偏振的色散关系:

$$\cos(K\Lambda) = \cos(k_{az}d_a)\cos(k_{bz}d_b) - \frac{1}{2}\left(\frac{k_1}{k_2}\frac{\mu_b}{\mu_a} + \frac{k_2}{k_1}\frac{\mu_a}{\mu_b}\right)\sin(k_{az}d_a)\sin(k_{bz}d_b),$$

$$\cos(K\Lambda) = \cos(k_{az}d_a)\cos(k_{bz}d_b) - \frac{1}{2}\left(\frac{k_1}{k_2}\frac{\epsilon_b}{\epsilon_a} + \frac{k_2}{k_1}\frac{\epsilon_a}{\epsilon_b}\right)\sin(k_{az}d_a)\sin(k_{bz}d_b).$$

Problem 117

导出 Lorentz 模型和 Drude 模型:

$$\epsilon_r(\omega) = 1 + \frac{\omega_p^2(\omega_0^2-\omega^2)}{(\omega_0^2-\omega^2)^2+\gamma^2\omega^2}, \quad \epsilon_r''(\omega) = \frac{\omega_p^2\gamma\omega}{(\omega_0^2-\omega^2)^2+\gamma^2\omega^2},$$

$$\epsilon_r(\omega) = 1 - \frac{\omega_p^2}{\omega^2 + \gamma^2}, \quad \epsilon_r''(\omega) = \frac{\omega_p^2 \gamma}{\omega(\omega^2 + \gamma^2)}.$$

Problem 118

导出电磁波在单轴晶体中传播的折射率椭球方程:

$$\frac{x^2}{n_0^2} + \frac{y^2}{n_0^2} + \frac{z^2}{n_e^2} = 1.$$

Problem 119

导出在旋光介质中传播的平面波满足的本征方程:

$$\begin{pmatrix} \epsilon & i\epsilon_g & 0 \\ i\epsilon_g & \epsilon & 0 \\ 0 & 0 & \epsilon_z \end{pmatrix} \vec{E} = n^2 \vec{E}.$$

Problem 120

导出电磁波在磁化等离子体中传播的介电张量:

$$\bar{\bar{\epsilon}} = \epsilon_0 \begin{pmatrix} \epsilon_1 & i\epsilon_2 & 0 \\ -i\epsilon_2 & \epsilon_1 & 0 \\ 0 & 0 & \epsilon_3 \end{pmatrix},$$

其中 $\epsilon_1 = 1 - \frac{\omega_p^2}{\omega^2 - \omega_c^2}$, $\epsilon_2 = \frac{\omega_c}{\omega} \frac{\omega_p^2}{\omega^2 - \omega_c^2}$.

Problem 121

导出双轴晶体中波矢面方程:

$$\frac{k_x^2}{n_1^{-2} - \omega^2/c^2} + \frac{k_y^2}{n_2^{-2} - \omega^2/c^2} + \frac{k_z^2}{n_3^{-2} - \omega^2/c^2} = 0.$$

Problem 122

证明电磁波在旋电介质中传播的色散关系:

$$\det\left(k^2\delta_{ij} - k_i k_j - \frac{\omega^2}{c^2}\mu_{ik}\epsilon_{kj}\right) = 0.$$

Problem 123

导出电磁波在双轴晶体中传播的两个可能的折射率:

$$n_\pm^2 = \frac{2}{\left(\frac{\sin^2\theta}{n_2^2} + \frac{\cos^2\theta}{n_3^2} + \frac{1}{n_1^2}\right) \pm \sqrt{\left(\frac{\sin^2\theta}{n_2^2} + \frac{\cos^2\theta}{n_3^2} + \frac{1}{n_1^2}\right)^2 + \frac{4\sin^2\theta\cos^2\theta}{n_2^2 n_3^2}}}.$$

Problem 124

导出电磁波在磁化铁氧体中传播的导磁张量:

$$\bar{\bar{\mu}} = \mu_0 \begin{pmatrix} \mu & i\kappa & 0 \\ -i\kappa & \mu & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

其中 $\mu = 1 + \frac{\omega_0\omega_m}{\omega_0^2 - \omega^2}$, $\kappa = \frac{\omega\omega_m}{\omega_0^2 - \omega^2}$.

Problem 125

导出电磁波在单轴晶体中传播的两个本征偏振态对应的折射率:

$$n_0 \quad 和 \quad \tilde{n}_e(\theta) = \left(\frac{\cos^2\theta}{n_0^2} + \frac{\sin^2\theta}{n_e^2} \right)^{-1/2}.$$

Problem 126

导出法拉第旋转角:

$$\theta = \mathcal{V} B d.$$

Problem 127

电偶极辐射功率:

$$P = \frac{\mu_0 \omega^4 p_0^2}{12\pi c}.$$

Problem 128

导出波导截止波数:

$$k_c = \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2}.$$

Problem 129

导出导体中衰减常数:

$$\alpha = \omega \sqrt{\frac{\mu\epsilon}{2}\left[\sqrt{1 + \left(\frac{\sigma}{\omega\epsilon}\right)^2} - 1\right]}.$$

Problem 130

证明电磁场不变量:

$$\mathcal{S} = \frac{1}{2}\left(\frac{E^2}{c^2} - B^2\right).$$

Problem 131

导出相对论变换下:

$$E'_\parallel = E_\parallel, \quad B'_\parallel = B_\parallel.$$

Problem 132

导出同步辐射功率:

$$P = \frac{2}{3}\frac{\mu_0 q^2 c\gamma^4 \omega_B^2}{\beta^2}.$$

Problem 133

证明辐射阻尼力:

$$\vec{F}_{\text{rad}} = \frac{\mu_0 q^2}{6\pi c}\dot{\vec{a}}.$$

Problem 134

导出李纳-维谢尔势:

$$A^\mu = \frac{\mu_0}{4\pi}\left[\frac{q u^\mu}{u\cdot(x-r)}\right]_{\text{ret}}.$$

Problem 135

导出运动电荷电场:

$$\vec{E} = \frac{q}{4\pi\epsilon_0}\frac{(1-\beta^2)(\vec{R}-R\vec{\beta})}{[R^2-(\vec{R}\times\vec{\beta})^2]^{3/2}}.$$

Problem 136

证明麦克斯韦方程组协变形式:

$$\partial_\mu F^{\mu\nu} = \mu_0 J^\nu.$$

Problem 137

推导矩形金属波导 ($a \times b$, $a > b$) 中主模 TE_{10} 的传输功率表达式:

$$P = \frac{ab}{4\eta}E_{10}^2\sqrt{1-\left(\frac{f_c}{f}\right)^2},$$

其中 $f_c = \dfrac{c}{2a}$, $\eta = \sqrt{\dfrac{\mu}{\epsilon}}$, E_{10} 为电场幅值.

Problem 138

导出圆柱形金属波导 (半径 R) 中 TM_{01} 模的截止频率:

$$f_c = \frac{x_{01}}{2\pi R\sqrt{\mu\epsilon}},$$

$x_{01} \approx 2.405$ 为 $J_0(x) = 0$ 的第一个根.

Problem 139

推导对称介质平板波导 (芯层厚 d, 折射率 n_1, 包层 n_2, $n_1 > n_2$) 中 TE_0 导模的色散方程:

$$\tan(\kappa d/2) = \frac{\gamma}{\kappa},$$

其中 $\kappa = \sqrt{k_0^2 n_1^2 - \beta^2}$, $\gamma = \sqrt{\beta^2 - k_0^2 n_2^2}$, β 为传播常数.

Problem 140

导出圆形介质波导 (光纤, 芯径 a, 折射率 n_1, 包层 n_2) 中 HE_{11} 模的特征方程:

$$\left[\frac{J_m'(u)}{uJ_m(u)} + \frac{K_m'(w)}{wK_m(w)}\right]\left[\frac{J_m'(u)}{uJ_m(u)} + \frac{n_2^2}{n_1^2}\frac{K_m'(w)}{wK_m(w)}\right] = m^2\left(\frac{1}{u^2} + \frac{1}{w^2}\right)\left(\frac{1}{u^2} + \frac{n_2^2}{n_1^2}\frac{1}{w^2}\right),$$

其中 $u = a\sqrt{k_0^2 n_1^2 - \beta^2}$, $w = a\sqrt{\beta^2 - k_0^2 n_2^2}$.

Problem 141

推导椭圆金属波导 (长半轴 a, 短半轴 b) 中主模 (TE_{c11}) 的截止波数近似解:

$$k_c \approx \frac{\pi}{2}\sqrt{\frac{1}{a^2} + \frac{1}{b^2}}.$$

Problem 142

导出脊形金属波导的等效截止波长近似公式:

$$\lambda_c \approx 2\left(a - t + \frac{2b}{\pi}\ln\left[\frac{1}{\sinh\left(\frac{\pi s}{2b}\right)}\right]\right),$$

其中 a, b, s, t 为脊形几何尺寸.

Problem 143

推导无耗传输线电报方程 $\dfrac{\partial^2 V}{\partial z^2} = LC\dfrac{\partial^2 V}{\partial t^2}$ 的通解，并给出特性阻抗 $Z_0 = \sqrt{\dfrac{L}{C}}$ 的表达式.

Problem 144

导出有耗传输线电报方程 $\dfrac{\partial^2 V}{\partial z^2} = (R{+}j\omega L)(G{+}j\omega C)V$, 传播常数 $\gamma = \sqrt{(R + j\omega L)(G + j\omega C)}$ 和特性阻抗 $Z_0 = \sqrt{\dfrac{R + j\omega L}{G + j\omega C}}$.

Problem 145

证明均匀传输线输入阻抗公式 $Z_{\rm in}(l) = Z_0\dfrac{Z_L + jZ_0\tan(\beta l)}{Z_0 + jZ_L\tan(\beta l)}$, 其中 $\beta = \omega\sqrt{LC}$ 为相位常数.

Problem 146

计算无限大正方形电阻网格 (每边电阻为 R) 中，坐标 (m, n) 与 $(0, 0)$ 两点间的等效电阻表达式 $R_{(m,n)}$, 满足二维差分方程 $4\phi_{m,n}-\phi_{m+1,n}-\phi_{m-1,n}-\phi_{m,n+1}-\phi_{m,n-1} = RI\delta_{m,0}\delta_{n,0}$.

Problem 147

推导无限长一维电阻网络 (相邻结点间电阻为 r) 上, 相隔 n 个正方形的结点之间的等效电阻

$$R = \frac{r}{2}\left[n + \frac{\sqrt{3}}{3}\left(1 + \left(2 - \sqrt{3}\right)^n\right)\right].$$

Problem 148

导出在无限大方形电阻网格中, 任意两格点间等效电阻

$$R = \frac{R}{4\pi} \iint_{\omega_{1,2}=0}^{\omega_{1,2}=2\pi} \frac{1-\cos(m\omega_1+n\omega_2)}{2-\cos(\omega_1)-\cos(\omega_2)} d\omega_1 d\omega_2.$$

.

Problem 149

证明 n 维有限长方体方体电阻网络 (棱边电阻均为 r) 中，坐标 $(x_1, x_2, \ldots, x_n)$ 到任意点 $(y_1, y_2, \ldots, y_n)$ 的等效电阻：

$$\begin{aligned}
R =& \frac{r}{\prod_{q=1}^{n} m_q} \sum_{k_1=0}^{m_1-1} \sum_{k_2=0}^{m_2-1} \cdots \sum_{k_n=0}^{m_n-1} \frac{2^{q-1}\left[\prod_{q=1}^{n} \cos\frac{2\pi k_q}{m_q}\left(x_q-\frac{1}{2}\right) - \prod_{q=1}^{n} \cos\frac{2\pi k_q}{m_q}\left(y_q-\frac{1}{2}\right)\right]}{n-\sum_{q=1}^{n}\cos\frac{2\pi k_q}{m_q}} \\
& \times \left[\cos\sum_{q=1}^{n}\frac{2\pi k_q}{m_q}\left(x_q-\frac{1}{2}\right) - \cos\sum_{q=1}^{n}\frac{2\pi k_q}{m_q}\left(y_q-\frac{1}{2}\right)\right] \\
=& \frac{r}{\prod_{q=1}^{n} m_q} \sum_{k_1=0}^{m_1-1} \sum_{k_2=0}^{m_2-1} \cdots \sum_{k_n=0}^{m_n-1} \frac{2^{q-1}\left[\prod_{q=1}^{n} \cos\frac{2\pi k_q}{m_q}\left(x_q-\frac{1}{2}\right) - \prod_{q=1}^{n} \cos\frac{2\pi k_q}{m_q}\left(y_q-\frac{1}{2}\right)\right]^2}{n-\sum_{q=1}^{n}\cos\frac{2\pi k_q}{m_q}}.
\end{aligned}$$

Problem 150

导出交流 RLC 串联电路的阻抗 $Z = R + j\left(\omega L - \dfrac{1}{\omega C}\right)$ 与谐振频率 $\omega_0 = \dfrac{1}{\sqrt{LC}}$, 并给出电流共振时的幅频特性曲线表达式.

Problem 151

证明在正弦稳态下, 任意线性无源单口网络的等效阻抗可表示为 $Z(j\omega) = R(\omega) + jX(\omega)$, 其中实部与虚部满足希尔伯特变换关系 (克拉默斯-克勒尼希关系).

Problem 152

推导三相平衡交流电路在对称负载下的线电压与相电压关系 $V_L = \sqrt{3}V_P e^{j\pi/6}$, 及复功率表达式 $\vec{S} = \sqrt{3}V_L I_L^*$.

Problem 153

导出包含互感的耦合谐振电路 (电感 L_1, L_2, 互感 M) 的系统方程, 并证明其简正模频率分裂为 $\omega = \omega_0/\sqrt{1 \pm k}$, 其中耦合系数 $k = M/\sqrt{L_1 L_2}$.

Problem 154

计算在直流激励下，半无限平面电阻网络 (边界为一条电阻链) 中，从边界点注入电流时, 网络内部任意点的电位分布 $\phi(x, y)$.

Problem 155

推导无耗、无畸变传输线 $(R/L = G/C)$ 的波动方程通解，并证明其特性阻抗为纯实数 $Z_0 = \sqrt{\dfrac{L}{C}}$.

Problem 156

导出周期加载传输线 (每节长度为 d, 并联导纳 Y) 的色散关系 $\cos(\beta d) = \cos(kd) - \dfrac{jYZ_0}{2}\sin(kd)$, 其中 $k = \omega\sqrt{LC}$, 并讨论其通带与禁带.

Problem 157

计算二维无限电阻网络在存在一个缺失电阻 (缺陷) 时, 原点与缺陷附近某点之间等效电阻的变化量 ΔR, 用网络格林函数表示。

Problem 158

证明对于由纯电阻构成的任意网络, 任意两结点 a, b 间的等效电阻 R_{ab} 可通过关联矩阵与电导矩阵表示为 $R_{ab} = (\mathbf{e}_a - \mathbf{e}_b)^T \mathbf{G}^+ (\mathbf{e}_a - \mathbf{e}_b)$, 其中 $\mathbf{G}^+$ 为电导矩阵的伪逆.

Problem 159

推导交流电路中, 非正弦周期信号 $v(t)$ 的有效值 $V_{\mathrm{rms}} = \sqrt{\frac{1}{T}\int_0^T v^2(t)dt}$ 及其傅里叶级数展开下的各次谐波功率叠加公式.

Part 4 光学

Problem 160

证明程函方程:

$$(\nabla S)^2 = n^2(\vec{r}).$$

Problem 161

证明光线方程:

$$\frac{d}{ds}\left(n\frac{d\vec{r}}{ds}\right) = \nabla n.$$

Problem 162

证明斯涅尔定律:

$$n_1 \sin\theta_1 = n_2 \sin\theta_2.$$

Problem 163

证明理想球面反射镜成像公式:

$$\frac{1}{u} + \frac{1}{v} = \frac{2}{R}.$$

Problem 164

证明薄透镜成像公式:

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}.$$

Problem 165

证明透镜横向放大率:

$$M = -\frac{v}{u}.$$

Problem 166

证明光在平行平板中产生的横向位移:

$$\Delta = d\sin\theta\left(1 - \frac{\cos\theta}{n\cos\theta'}\right).$$

Problem 167

证明在平方律介质 $n^2(x) = n_0^2(1-\alpha^2x^2)$ 中, 近轴光线轨迹为

$$x(z) = x_0\cos(\alpha z) + \frac{\theta_0}{\alpha}\sin(\alpha z).$$

Problem 168

证明在球对称分布 $n(r) = n_0/r$ 中, 光线轨迹为圆

$$r = R_0\sec(\theta-\theta_0).$$

Problem 169

证明在折射率指数分布 $n(y) = n_0e^{\beta y}$ 中, 光线轨迹为抛物线

$$y(z) = y_0 + \theta_0 z + \frac{\beta z^2}{2}.$$

Problem 170

证明在轴向渐变光纤 $n^2(r) = n_0^2[1 - 2\Delta(r/a)^g]$ 中, 当 $g = 2$ 时子午光线满足

$$\frac{d^2r}{dz^2} + \left(\frac{2\Delta}{a^2 n_0^2}\right) r = 0.$$

Problem 171

证明平移矩阵：

$$\mathbf{M} = \begin{pmatrix} 1 & 0 \\ \dfrac{d}{n} & 1 \end{pmatrix}.$$

Problem 172

证明折射矩阵：

$$\mathbf{M} = \begin{pmatrix} 1 & \dfrac{n_1 - n_2}{r} \\ 0 & 1 \end{pmatrix}.$$

Problem 173

证明反射矩阵：

$$\mathbf{M} = \begin{pmatrix} 1 & -\dfrac{2n}{r} \\ 0 & 1 \end{pmatrix}.$$

Problem 174

证明薄透镜矩阵：

$$\mathbf{M} = \begin{pmatrix} 1 & -\dfrac{1}{f} \\ 0 & 1 \end{pmatrix}.$$

Problem 175

证明在周期透镜序列 (焦距 f, 间距 d) 中, 一个周期的传输矩阵为:

$$\mathbf{M} = \begin{pmatrix} 1 - d/f & d \\ -1/f & 1 \end{pmatrix}.$$

Problem 176

证明在渐变折射率介质 ($n^2(x) = n_0^2 - \alpha x^2$) 中, 长度为 L 的传输矩阵为

$$\mathbf{M} = \begin{pmatrix} \cos(\alpha L) & \sin(\alpha L)/\alpha \\ -\alpha \sin(\alpha L) & \cos(\alpha L) \end{pmatrix}.$$

Problem 177

证明在非均匀介质中, 光线传输的普遍矩阵形式为:

$$\begin{pmatrix} x_1 \\ \theta_1 \end{pmatrix} = \begin{pmatrix} C(z) & S(z) \\ C'(z) & S'(z) \end{pmatrix} \begin{pmatrix} x_0 \\ \theta_0 \end{pmatrix}.$$

Problem 178

证明拉格朗日-亥姆霍兹不变量:

$$L = ny\alpha.$$

Problem 179

证明傍轴近似下高斯光束解满足:

$$w(z) = w_0\sqrt{1 + (z/z_R)^2}.$$

Problem 180

证明下述曲面可使宽光束严格成像

$$\left(n_2^2-n_1^2\right)z^2-2\left(n_2-n_1\right)n_2fz+n_2^2y^2=0.$$

Problem 181

证明阿贝正弦条件:

$$ny\sin u=\text{const}.$$

Problem 182

证明望远镜中:

$$F=\frac{f_of_e}{f_o+f_e-d},u_h=-\frac{f_od}{f_o+f_e-d},v_h=\frac{f_ed}{f_o+f_e-d}.$$

并分伽利略望远镜，开普勒望远镜讨论 F,u_h,v_h 的符号.

Problem 183

证明显微镜中:

$$M_{eff}=2\text{N.A.}\frac{l_0}{D_e}.$$

Problem 184

证明薄透镜相位变换因子:

$$t_l = \exp\left(-i\frac{k}{2f}(x^2+y^2)\right).$$

Problem 185

证明菲涅尔衍射积分:

$$U(x,y) = \frac{e^{ikz}}{i\lambda z}\int U_0(\xi,\eta)e^{\frac{ik}{2z}[(x-\xi)^2+(y-\eta)^2]}d\xi d\eta.$$

Problem 186

证明夫琅禾费衍射是孔径的傅里叶变换:

$$U(x,y) \propto \mathcal{F}\{U_0(\xi,\eta)\}.$$

Problem 187

证明单缝衍射光强:

$$I(\theta) = I_0\left[\frac{\sin(\pi a\sin\theta/\lambda)}{\pi a\sin\theta/\lambda}\right]^2.$$

Problem 188

证明圆孔衍射艾里斑光强:

$$I(r) = I_0\left[\frac{2J_1(kar/z)}{(kar/z)}\right]^2.$$

Problem 189

证明菲涅尔波带片:

$$\frac{1}{a}+\frac{1}{b}=(2n+1)\,\frac{\lambda}{\rho_1^2}.$$

平面波入射出射光场：

$$U_2\left(r\right)=\frac{A}{2}+\frac{A}{\pi i}\sum_{n=1}^{\infty}\frac{1}{2n+1}\left[e^{i(2n+1)\pi r^2/\rho_1^2}-e^{-i(2n+1)\pi r^2/\rho_1^2}\right].$$

Problem 190

证明单缝菲涅尔衍射振幅：

$$U\left(\vec{r}\right)=\frac{(1-i)\,Ae^{ik(z+z_0)}}{2\left(z+z_0\right)}\exp\left[\frac{ik}{2}\frac{\left(x-x_0\right)^2+y^2}{z+z_0}\right]\int_{w_2}^{w_1}e^{i\dfrac{\pi u^2}{2}}du.$$

Problem 191

证明双光束干涉光强分布:

$$I=4I_0\cos^2\left(\frac{\delta}{2}\right),$$

其中 $\delta=\dfrac{2\pi}{\lambda}\Delta L.$

Problem 192

证明法布里-珀罗干涉仪透射峰位满足:

$$2nd\cos\theta=m\lambda.$$

Problem 193

证明牛顿环暗环半径:

$$r_m = \sqrt{m\lambda R}.$$

Problem 194

证明迈克尔逊干涉仪中, 动镜移动 Δd 引起条纹移动数

$$\Delta m = \frac{2\Delta d}{\lambda}.$$

Problem 195

证明多缝光栅光强:

$$I(\theta) = I_0 \left(\frac{\sin\alpha}{\alpha}\right)^2 \left(\frac{\sin N\beta}{\sin\beta}\right)^2,$$

其中 $\alpha = \dfrac{\pi a \sin\theta}{\lambda}$, $\beta = \dfrac{\pi d \sin\theta}{\lambda}$.

Problem 196

证明光栅方程:

$$d(\sin\theta_m \pm \sin\theta_i) = m\lambda.$$

Problem 197

证明光栅角色散:

$$\frac{d\theta}{d\lambda} = \frac{m}{d\cos\theta_m}.$$

Problem 198

证明光栅分辨率:

$$R = \frac{\lambda}{\Delta\lambda} = mN.$$

Problem 199

证明闪耀光栅条件:

$$2d \sin\theta_B = m\lambda.$$

Problem 200

证明正弦振幅光栅的衍射效率:

$$\eta = \left[\frac{J_1(\phi_m/2)}{\phi_m/4}\right]^2.$$

Problem 201

证明瑞利判据下, 最小可分辨角距离

$$\theta_{\min} = 1.22\frac{\lambda}{D}.$$

Problem 202

证明光学传递函数:

$$\mathcal{H}(f_x, f_y) = \mathcal{F}\{h(x, y)\}.$$

Problem 203

证明马吕斯定律:

$$I = I_0 \cos^2 \theta.$$

Problem 204

证明任意偏振态可用斯托克斯参量 (S_0, S_1, S_2, S_3) 描述, 且满足

$$S_0^2 \geq S_1^2 + S_2^2 + S_3^2.$$

Problem 205

证明线偏振器的穆勒矩阵为:

$$\frac{1}{2}\begin{pmatrix} 1 & \cos 2\theta & \sin 2\theta & 0 \\ \cos 2\theta & \cos^2 2\theta & \sin 2\theta \cos 2\theta & 0 \\ \sin 2\theta & \sin 2\theta \cos 2\theta & \sin^2 2\theta & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Problem 206

证明波片 (相位延迟 δ, 快轴方位角 $0°$) 的穆勒矩阵为

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cos\delta & \sin\delta \\ 0 & 0 & -\sin\delta & \cos\delta \end{pmatrix}.$$

Problem 207

证明部分偏振光偏振度:

$$P = \frac{\sqrt{S_1^2 + S_2^2 + S_3^2}}{S_0}.$$

Problem 208

证明椭圆偏振光的方位角 ψ 与椭圆率角 χ 满足:

$$\tan 2\psi = \frac{S_2}{S_1}, \quad \sin 2\chi = \frac{S_3}{S_0}.$$

Problem 209

证明在考虑偏振效应后, 薄膜的透射率公式应修正为:

$$T = \frac{T_s + T_p}{2}.$$

Problem 210

证明朗伯辐射体的辐射亮度 L_e 为常数, 其辐射强度角分布满足:

$$I_e(\theta) = I_0 \cos\theta.$$

Problem 211

证明点光源在距离 R 处产生的照度 E_v 满足平方反比定律:

$$E_v = \frac{I_v}{R^2}.$$

Problem 212

证明发光强度的余弦三次方定律: 面元 dA 在与其法线成 θ 角方向上的照度:

$$dE = \frac{I_0 \cos^3 \theta}{R^2} dA.$$

Problem 213

证明准单色光的时间相干长度:

$$L_c = \frac{\lambda_0^2}{\Delta\lambda}.$$

Problem 214

证明在非相干光照明下, 两个空间部分相干点源的干涉条纹可见度:

$$V = |\gamma_{12}(\tau)|.$$

Problem 215

证明扩展光源产生的干涉条纹可见度与光源尺寸满足:

$$V = \left| \operatorname{sinc}\left(\frac{\pi b \xi}{\lambda z}\right) \right|.$$

Problem 216

证明部分相干光的互强度 $J_{12}(\tau)$ 与互相干函数 $\Gamma_{12}(\tau)$ 满足:

$$\Gamma_{12}(\tau) = \langle U_1(t+\tau) U_2^*(t) \rangle.$$

Problem 217

证明范西特-泽尼克定理: 扩展准单色光源在观察面上产生的复相干度:

$$\mu_{12} = \frac{\iint_S I(\xi,\eta)e^{ik(\Delta R)}d\xi d\eta}{\iint_S I(\xi,\eta)d\xi d\eta}.$$

Problem 218

证明在迈克尔逊测星干涉仪中, 条纹可见度与恒星角直径 α 的关系为:

$$V = \left|\frac{2J_1\left(\dfrac{\pi B\alpha}{\lambda}\right)}{\dfrac{\pi B\alpha}{\lambda}}\right|.$$

Problem 219

证明两束部分相干光叠加后的光强:

$$I = I_1 + I_2 + 2\sqrt{I_1I_2}|\gamma_{12}(\tau)|\cos[\alpha_{12}(\tau)].$$

Problem 220

证明高斯-谢尔模型光束的横向相干长度:

$$\rho_0 = \frac{\lambda}{\pi\theta_s}.$$

Problem 221

证明在二阶非线性过程中, 倍频效率:

$$\eta \propto \frac{d_{\text{eff}}^2 L^2 I_\omega}{n^3\lambda^2}\text{sinc}^2\left(\frac{\Delta kL}{2}\right).$$

Problem 222

证明光在非线性介质中耦合波方程:

$$\frac{dA_1}{dz} = i\kappa A_2 A_3^* e^{i\Delta kz}.$$

Part 5 近代物理

Problem 223

证明理想 pn 结在正偏压 V 下的电流密度满足

$$J = J_s \left(e^{eV/(k_B T)} - 1\right).$$

Problem 224

证明两平行理想导体板间的卡西米尔力为

$$F = -\frac{\pi^2 \hbar c A}{240 d^4},$$

其中 A 为板面积, d 为间距.

Problem 225

证明在椭圆轨道修正的玻尔模型中, 氢原子能级仍为

$$E_n = -\frac{m e^4}{2(4\pi\epsilon_0)^2 \hbar^2 n^2}.$$

Problem 226

证明考虑相对论修正的玻尔模型能级为

$$E_{n,j} = -\frac{mc^2\alpha^2}{2n^2}\left[1 + \frac{\alpha^2}{n}\left(\frac{1}{j+\frac{1}{2}} - \frac{3}{4n}\right)\right],$$

其中 α 为精细结构常数.

Problem 227

证明康普顿散射中光子波长偏移

$$\Delta\lambda = \frac{h}{m_e c}(1 - \cos\theta).$$

Problem 228

证明相对论性前灯效应中, 辐射角分布满足

$$\frac{dP}{d\Omega} \propto \frac{1}{\gamma^4(1 - \beta\cos\theta)^4}.$$

Problem 229

证明经典霍尔效应中霍尔电压

$$V_H = \frac{IB}{net},$$

其中 n 为载流子浓度, t 为样品厚度.

Problem 230

证明在均匀磁场 B 中, 电子的朗道能级为

$$E_n = \hbar\omega_c\left(n + \frac{1}{2}\right),$$

其中 $\omega_c = \dfrac{eB}{m}$.

Problem 231

证明量子霍尔效应中霍尔电阻

$$R_H = \frac{h}{\nu e^2},$$

ν 为填充数.

Problem 232

证明一维无限深方势阱中粒子的能级

$$E_n = \frac{n^2\pi^2\hbar^2}{2mL^2}.$$

Problem 233

证明谐振子能级

$$E_n = \hbar\omega\left(n + \frac{1}{2}\right).$$

Problem 234

证明氢原子基态波函数

$$\psi_{100} = \frac{1}{\sqrt{\pi a_0^3}} e^{-r/a_0},$$

其中 a_0 为玻尔半径.

Problem 235

证明不确定关系

$$\Delta x \Delta p \geq \frac{\hbar}{2}.$$

Problem 236

证明泡利矩阵满足对易关系

$$[\sigma_i, \sigma_j] = 2i\epsilon_{ijk}\sigma_k.$$

Problem 237

证明隧道效应中, 方势垒的透射系数

$$T \approx e^{-2a\sqrt{2m(V_0-E)}/\hbar}.$$

Problem 238

证明在周期 δ 势 $V(x) = \alpha \sum_{n=-\infty}^{\infty} \delta(x - na)$ 中, 电子波函数满足的边界条件为

$$\psi'(0^+) - \psi'(0^-) = \frac{2m\alpha}{\hbar^2}\psi(0).$$

Problem 239

证明克朗尼格-潘纳模型的能带方程

$$\cos(ka) = \cos(qa) + \frac{m\alpha}{\hbar^2 q}\sin(qa),$$

其中 $q = \sqrt{2mE}/\hbar$.

Problem 240

证明在紧束缚近似下, 一维单原子链的能带为

$$E(k) = E_0 - 2J\cos(ka),$$

其中 J 为交叠积分.

Problem 241

证明在近自由电子近似下, 一维晶格在布里渊区边界 $k = \pm\pi/a$ 处出现能隙

$$\Delta = 2|V_G|,$$

其中 V_G 为相应倒格矢的傅里叶分量.

Problem 242

证明布洛赫定理

$$\psi_{n\vec{k}}(\vec{r} + \vec{R}) = e^{i\vec{k}\cdot\vec{R}}\psi_{n\vec{k}}(\vec{r}).$$

Problem 243

证明电子有效质量 m^* 满足

$$\frac{1}{m^*} = \frac{1}{\hbar^2}\frac{d^2E}{dk^2}.$$

Problem 244

证明在德拜模型下, 固体低温比热

$$C_V \propto T^3.$$

Problem 245

证明自由电子的费米能

$$E_F = \frac{\hbar^2}{2m}(3\pi^2 n)^{2/3}.$$

Problem 246

证明自由粒子的路径积分传播子

$$K(x_b, t_b; x_a, t_a) = \sqrt{\frac{m}{2\pi i\hbar(t_b - t_a)}} \exp\left[\frac{im(x_b - x_a)^2}{2\hbar(t_b - t_a)}\right].$$

Problem 247

证明谐振子的路径积分传播子

$$K(x_b, t_b; x_a, t_a) = \sqrt{\frac{m\omega}{2\pi i\hbar \sin\omega T}} \exp\left\{\frac{im\omega}{2\hbar\sin\omega T}\left[(x_a^2 + x_b^2)\cos\omega T - 2x_a x_b\right]\right\},$$

$s.t.\, T = t_b - t_a$.

Problem 248

证明路径积分表述下的跃迁振幅

$$\langle x_b | e^{-i\hat{H}t/\hbar} | x_a \rangle = \int \mathcal{D}[x(t)]\, e^{\frac{i}{\hbar}S[x(t)]}.$$

Problem 249

证明在经典路径附近展开, 作用量可写为

$$S[x(t)] = S_{\rm cl} + \delta^2 S + \cdots .$$

Problem 250

证明瞬子解在双势阱隧穿计算中给出的能级分裂

$$\Delta E \propto e^{-S_0/\hbar},$$

其中 S_0 为欧氏作用量.

Problem 251

证明相对论性多普勒效应频率公式

$$\nu' = \nu\sqrt{\frac{1-\beta}{1+\beta}},$$

其中 $\beta = v/c$, 光源与观察者相背运动.

Problem 252

证明相对论性粒子的拉格朗日量

$$L = -m_0c^2\sqrt{1-\beta^2}.$$

Problem 253

证明核反应 $A(a,b)B$ 的反应能

$$Q = (m_A + m_a - m_B - m_b)c^2.$$

Problem 254

证明放射性衰变规律

$$N(t) = N_0 e^{-\lambda t},$$

其中 λ 为衰变常数.

Problem 255

证明半衰期 $T_{1/2}$ 与衰变常数关系

$$T_{1/2} = \frac{\ln 2}{\lambda}.$$

Problem 256

证明 α 衰变中, 衰变常数 λ 与伽莫夫因子 G 满足

$$\lambda \propto e^{-G},$$

其中 $G = \dfrac{2\pi Z_1 Z_2 e^2}{\hbar v}$.

Problem 257

证明核反应阈能

$$E_{\text{th}} = \frac{m_A + m_a}{m_A}|Q|,$$

当 $Q < 0$ 时.

Problem 258

证明结合能

$$B = [Zm_p + (A - Z)m_n - M_{\text{核}}]c^2.$$

Problem 259

证明液滴模型半经验质量公式

$$B(A,Z)=a_VA-a_SA^{2/3}-a_C\frac{Z^2}{A^{1/3}}-a_A\frac{(A-2Z)^2}{A}+\delta(A,Z).$$

Problem 260

证明在液滴模型下, 最稳定原子核的 Z 满足

$$\frac{Z}{A}=\frac{1}{2+\dfrac{a_C}{2a_A}A^{2/3}}.$$

Problem 261

证明壳模型中, 核子数 $2,8,20,28,50,82,126$ 为幻数.

Problem 262

证明集体模型 (几何模型) 中, 原子核四极形变参数 β 与电四极矩 Q 的关系

$$Q=\frac{3}{\sqrt{5\pi}}ZR_0^2\beta.$$

Problem 263

证明费米气体模型下, 原子核的能级密度

$$\rho(E)\propto\exp(2\sqrt{aE}),$$

其中 a 为能级密度参数.

Problem 264

证明在独立粒子模型下, 原子核磁矩的施密特值

$$\mu = g_j j,$$

其中 g_j 为朗德 g 因子.